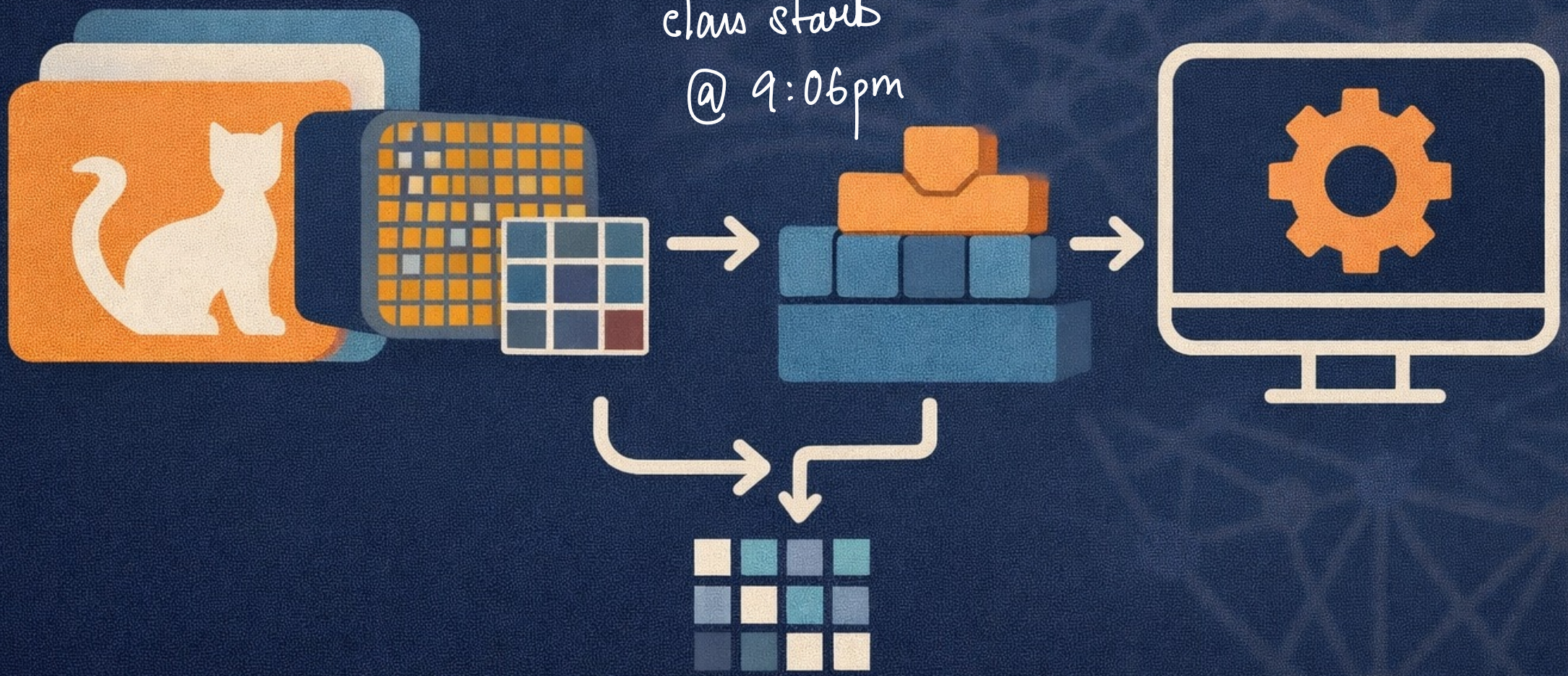
CONVOLUTIONAL NEURAL NETWORK



Lecture : Resnet and Object detection

#Agenda

1) Resnet

2) Object detection

Recap

1) Image similarity problem

Conv → Embedding

↓

Dense

Image → Embedding

$(I_1) \rightarrow (E_1)$

↳ Query image

Cosine Sim, Euclidean dist, manhatten dist, KNN

2) Inception Net

↳ google

(Inception block) → Repeating

9 times

"Deeper"

"Vanishing gradient / Exploding gradient"

Resnet

↳ Residual Network

↳ leftover

"Skip Connection"

Image Net

AlexNet

VGG

InceptionNet

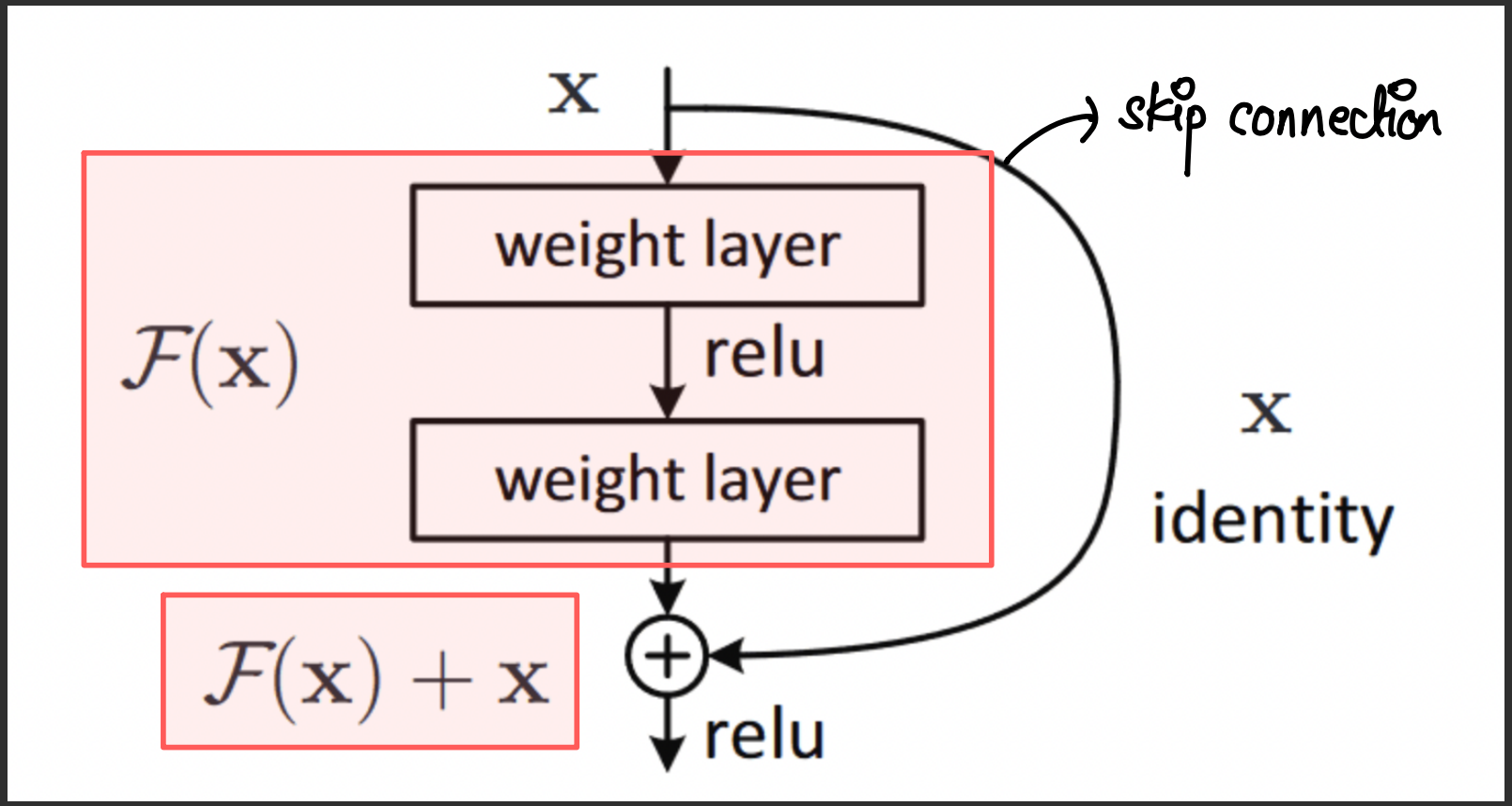
Resnet

↳ MS labs

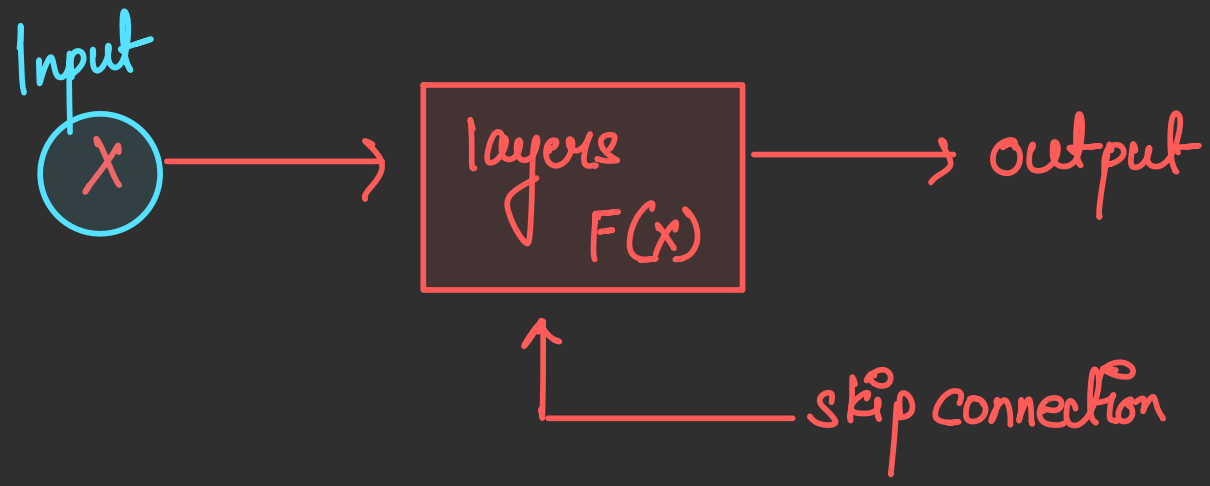
X → layer 1 → layer 2 → layer 3 → output

$W_1 = 0.23$

output → H(x)



Output = F(x) + X



$x = 6$

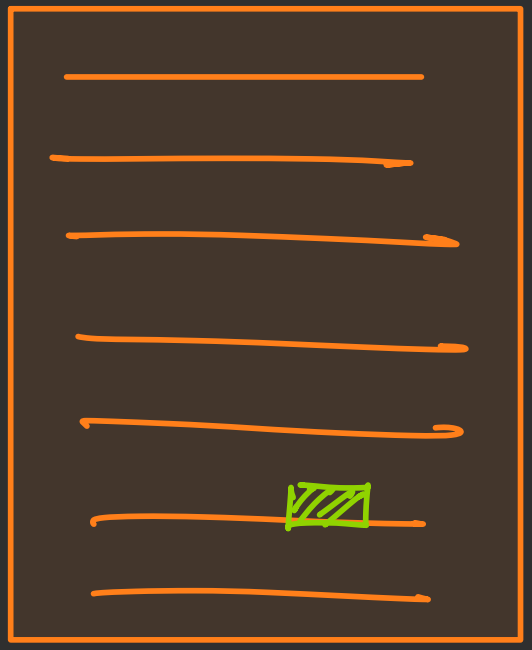
$H(x) = 4$

$F(x) = 2$

min

$$\text{output}\,(H(x)) = F(x) + x$$

$$F(x) = H(x) - x$$



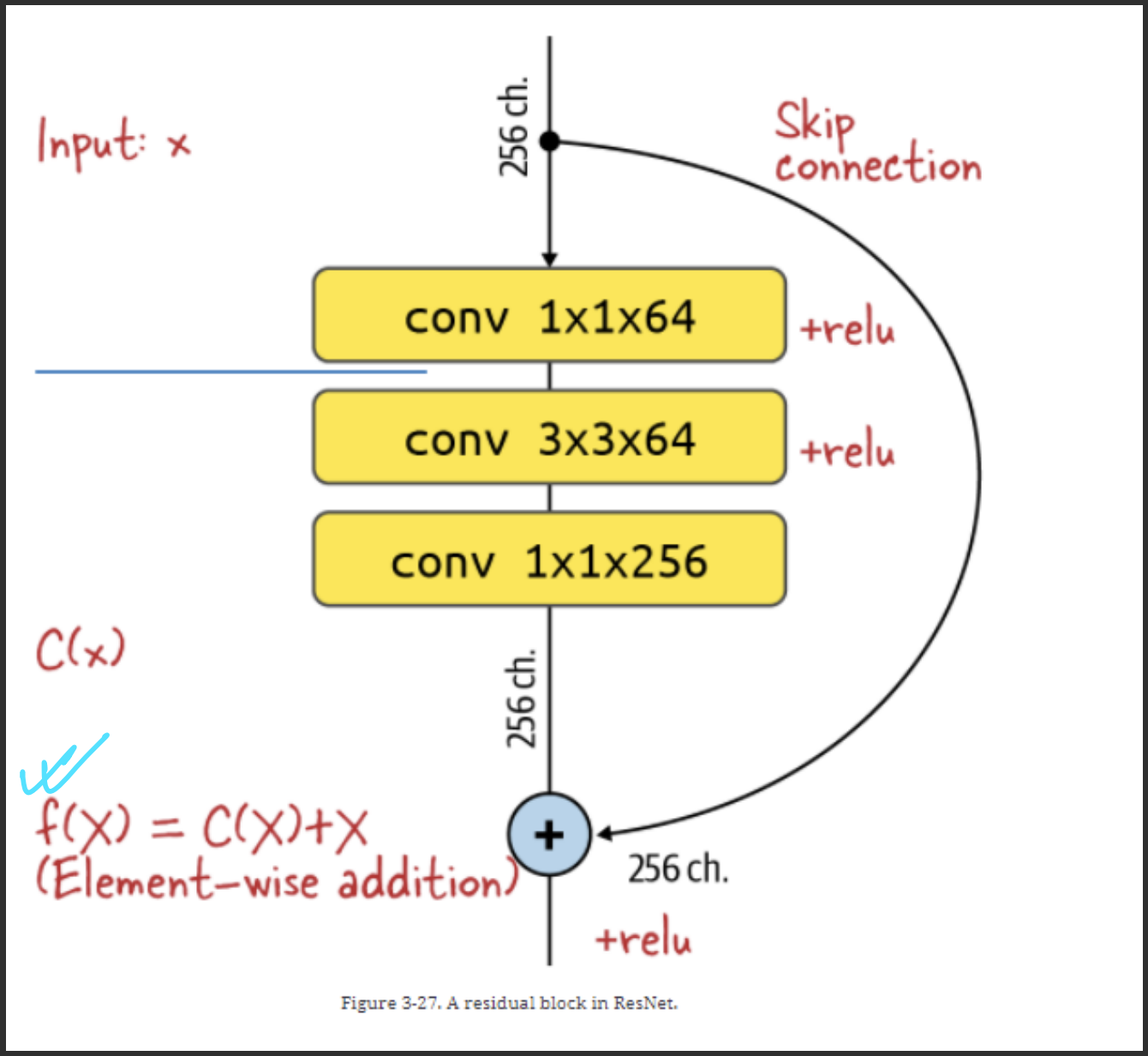
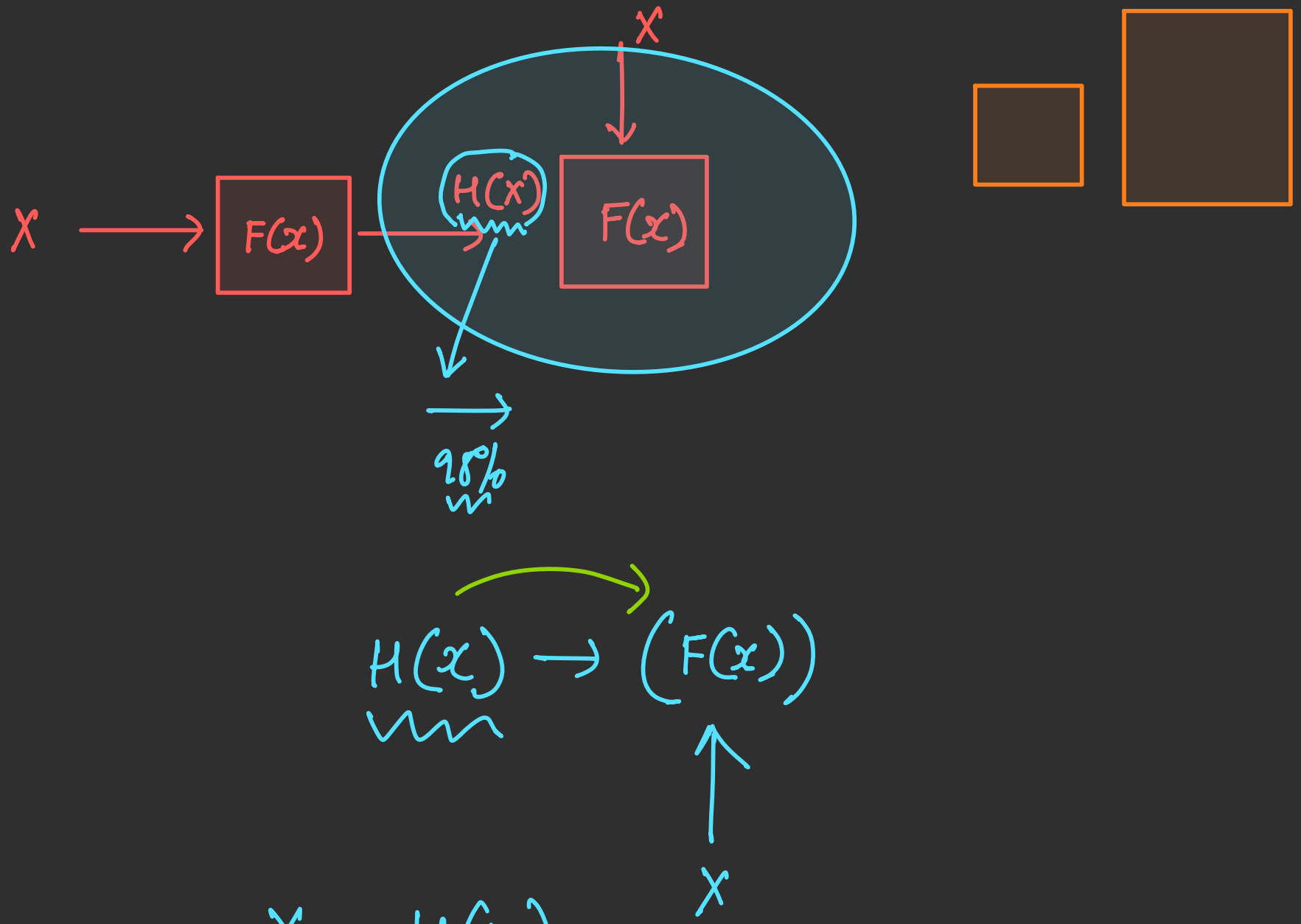


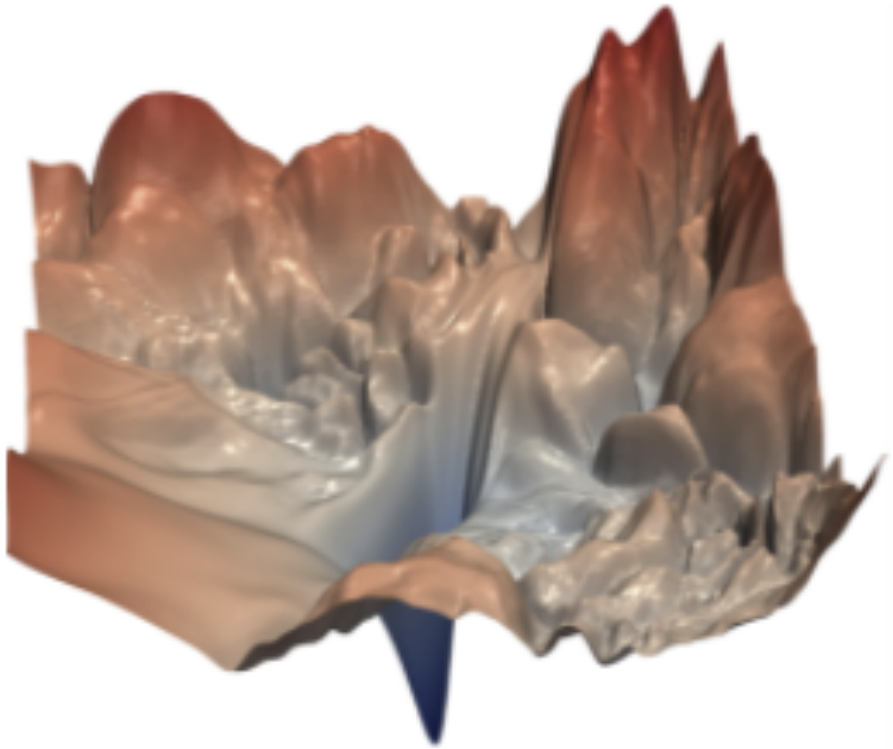
Figure 3-27. A residual block in ResNet.

$x \rightarrow$ Input

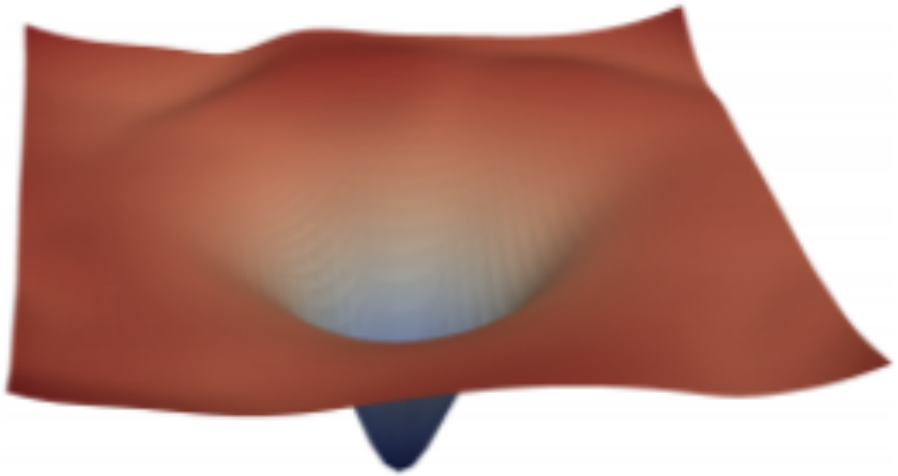
$C(x) \rightarrow$ Learn something

$f(x) = C(x) + x$

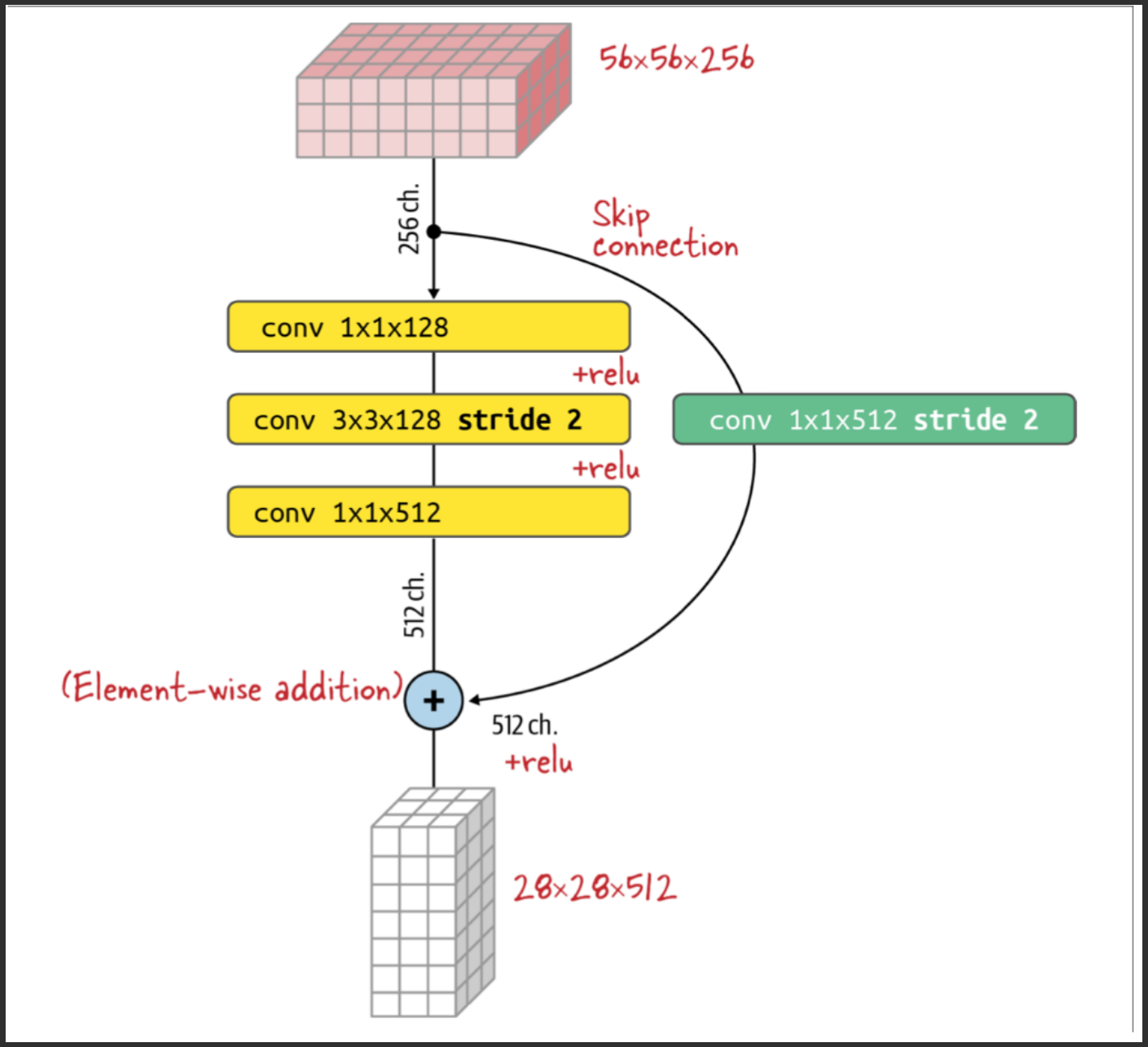

X
F(x)
H(x)
F(x)
X
98%
H(x) → (F(x))
X
X — H(x)



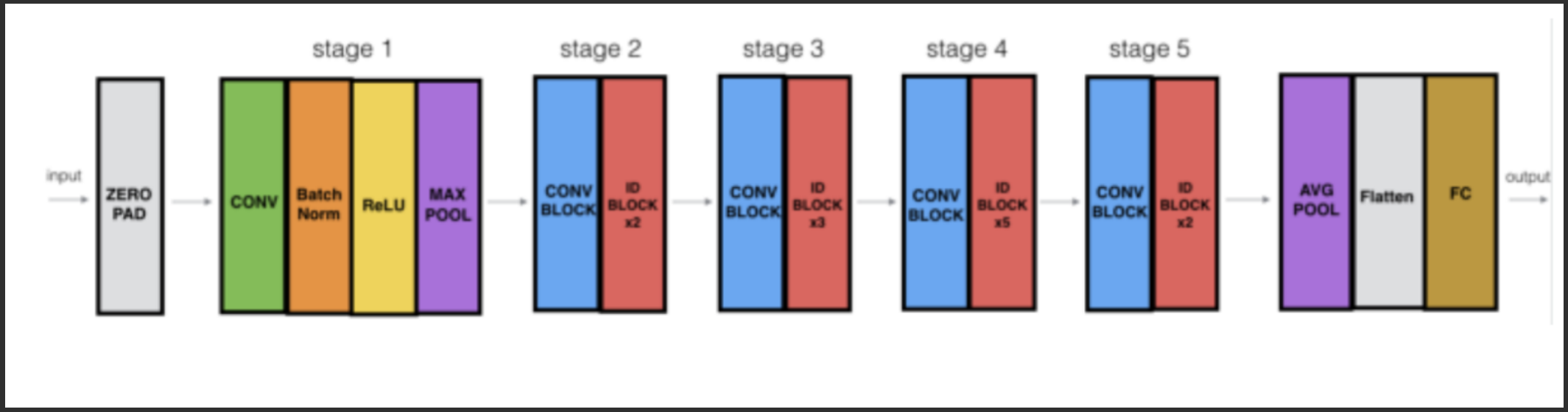
(a) without skip connections

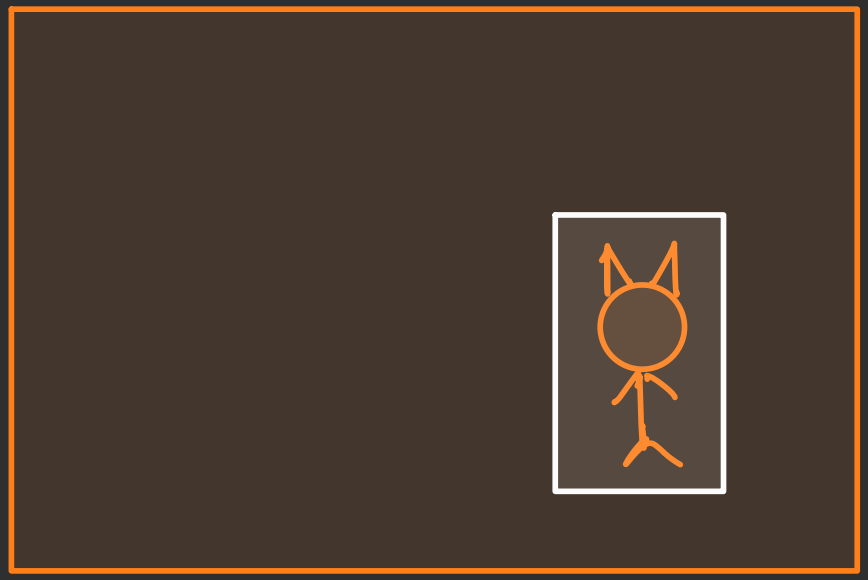


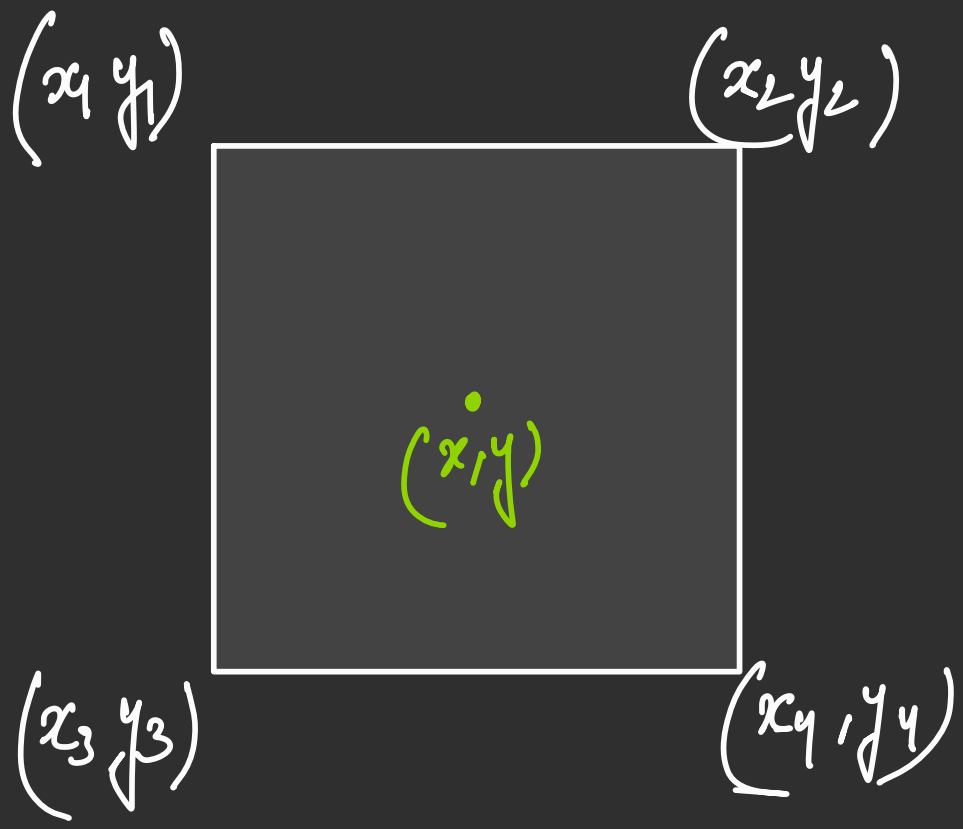
(b) with skip connections

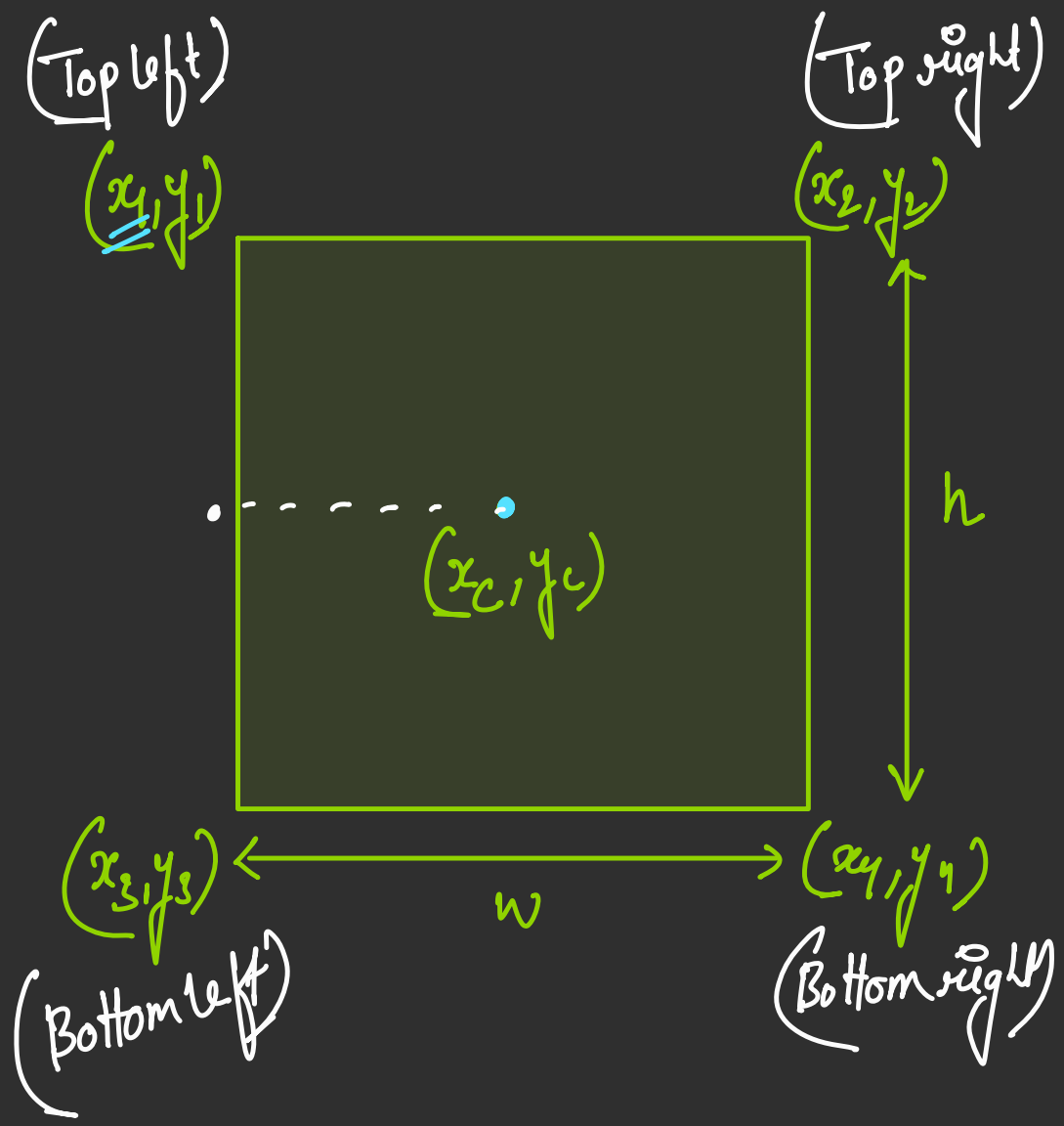

56×56×256
256 ch.
Skip connection
conv 1x1x128
+relu
conv 3x3x128 stride 2
+relu
conv 1x1x512
conv 1x1x512 stride 2
512 ch.
(Element-wise addition)
512 ch.
+relu
28×28×512

Resnet Architecture





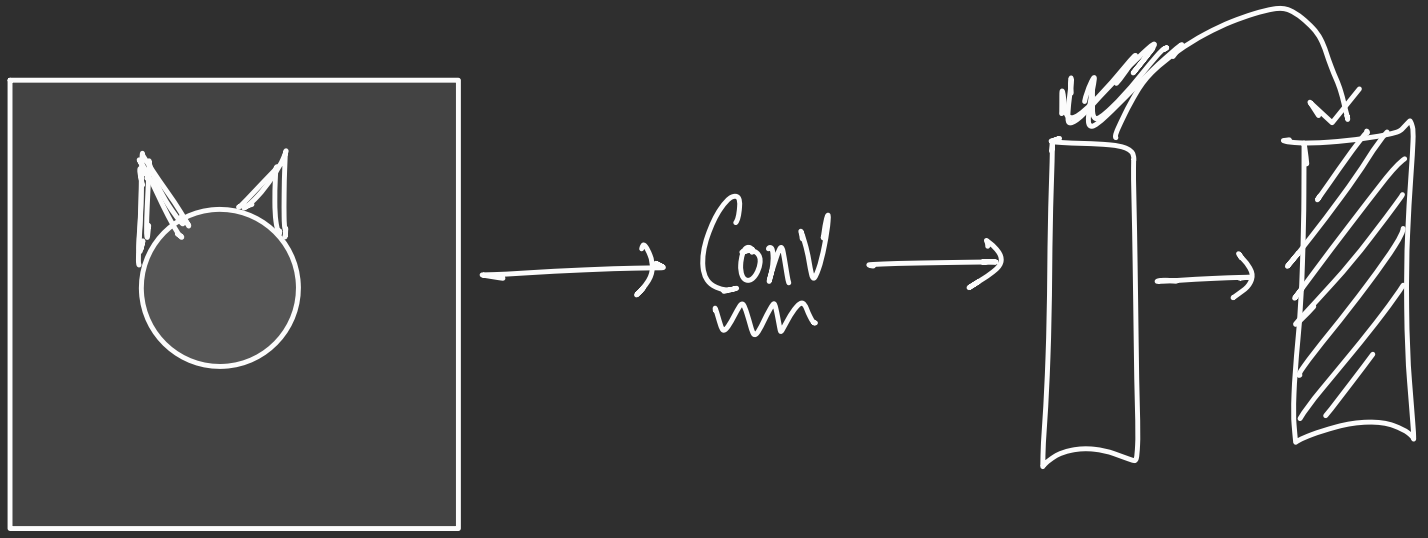

(x_1, y_1)
(x_2, y_2)
(x, y)
(x_3, y_3)
(x_4, y_4)

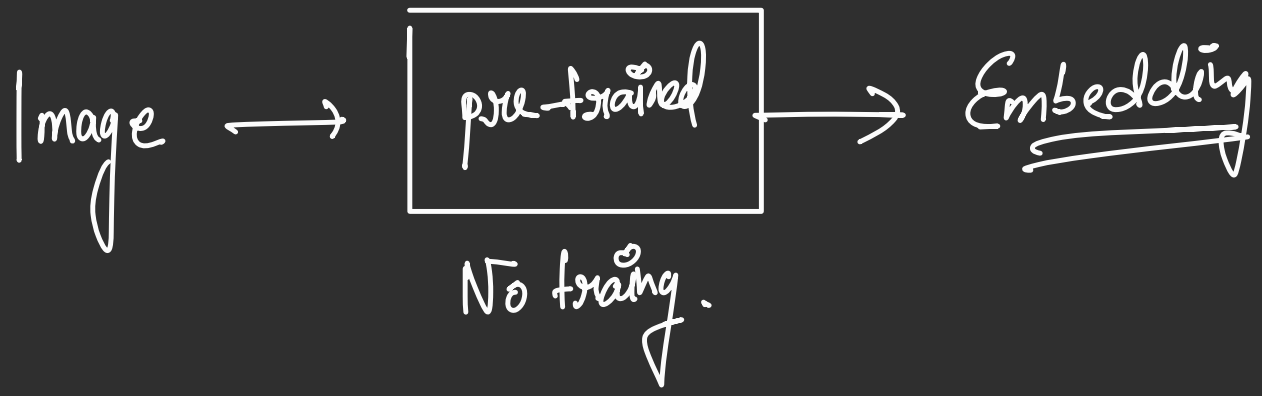


- x_{min}, y_{min} ✓
- x_{max}, y_{min} ✓
- x_{max}, y_{max} ✓
- x_{min}, y_{max} ✓

$$x_{min} = x_{center} - \frac{width}{2}$$

$$x_{max} = x_{center} + \frac{width}{2}$$


Conv



model

model.predict(X_train)

Embedding